ROBOTIS OPENMANIPULATOR-X · 4-DOF + GRIPPER · XM430-W350 IDS 11–15

Joint frames, positive directions and limits

Every frame, arrow and dimension below is drawn to scale from the URDF that ROBOTIS ships (`open_manipulator_x_arm.urdf.xacro`) and the dimension drawing on the ROBOTIS e-Manual specification page. Angles are the values on `/joint_states`, in radians, with degrees alongside.

All frames start parallel

Every joint origin has rpy = 0, so at the zero pose each link frame points x forward, y left, z up, the same as world.

Right-hand rule

Thumb along the joint axis, fingers curl the positive way. joint1 turns about +z; joints 2, 3, 4 all turn about +y.

Positive = down for 2, 3, 4

Turning about +y moves +z toward +x, so a positive θ2, θ3 or θ4 tips that link forward and down.

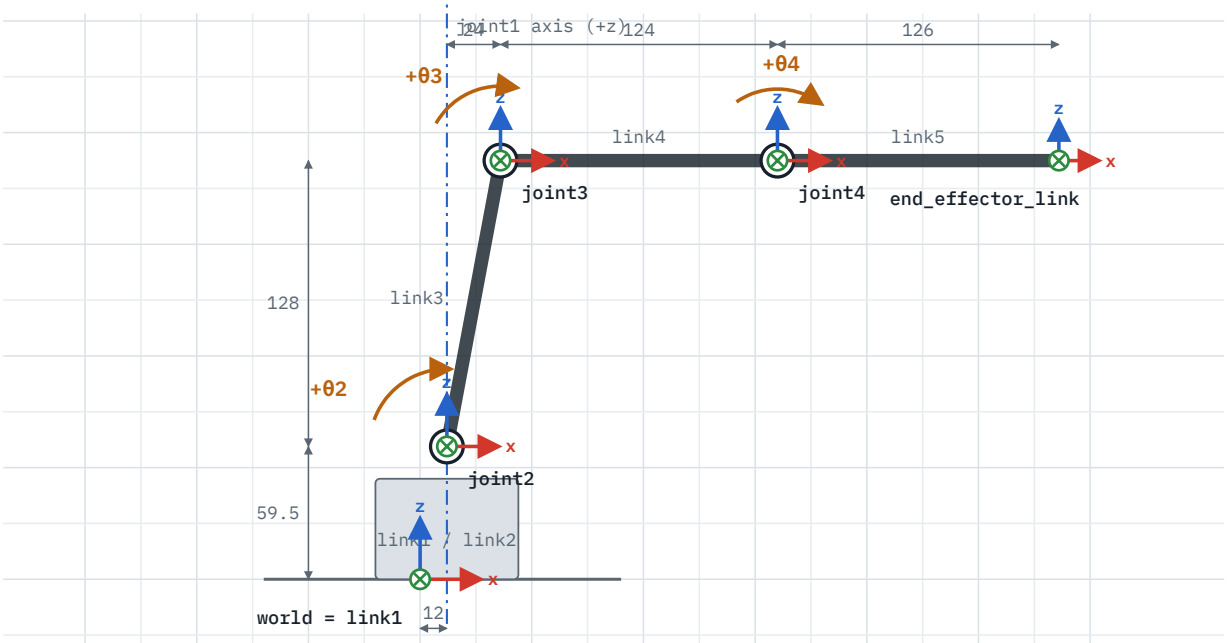
Pitches add up

The three axes are parallel, so gripper pitch below horizontal = θ2 + θ3 + θ4 (what `ee_commander` calls pitch).

FIGURE 1 · SIDE VIEW (X–Z PLANE)

Zero | Home | Exit

Joint frames and positive rotations of joints 2, 3, 4

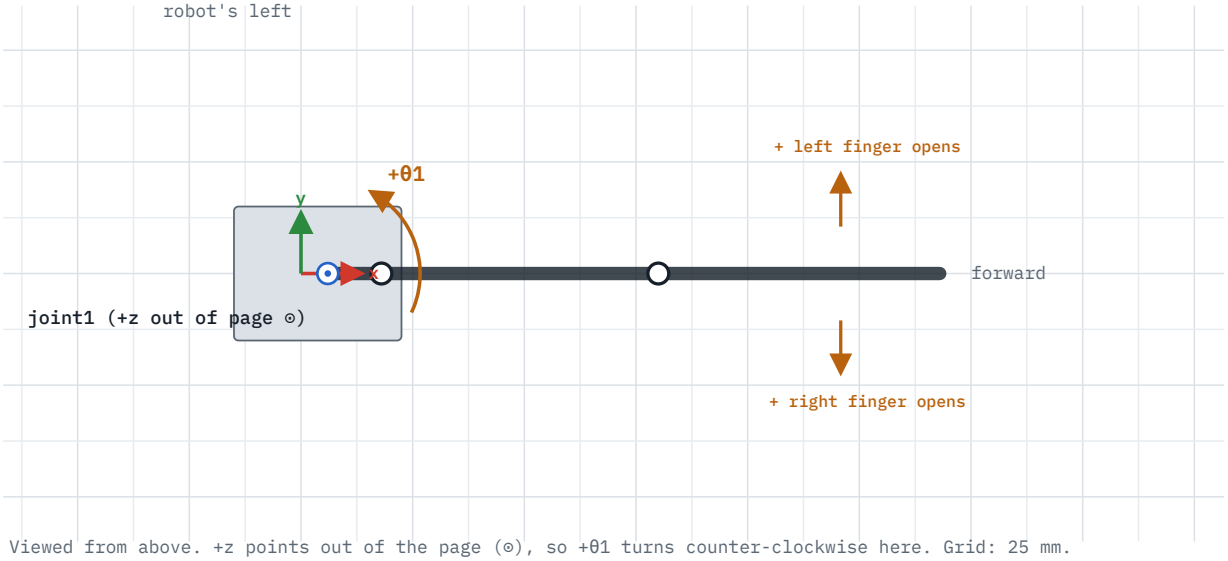


Viewed from the robot's right side. +y points into the page (⊗), so every positive θ2, θ3, θ4 turns clockwise here
Grid: 25 mm (bold every 50 mm). Joints, links and frames to scale; base block schematic.

Zero pose [0, 0, 0, 0]: the "L". All joint frames are parallel to world (x forward, y left, z up).

FIGURE 2 · TOP VIEW (X–Y PLANE)

Joint1 rotation and gripper opening direction



Viewed from above. +z points out of the page (⊙), so +θ1 turns counter-clockwise here. Grid: 25 mm.

The pose buttons above also move this view. In the home and exit poses joint1 is turned +34.9° and +26.5°, toward the robot's left.

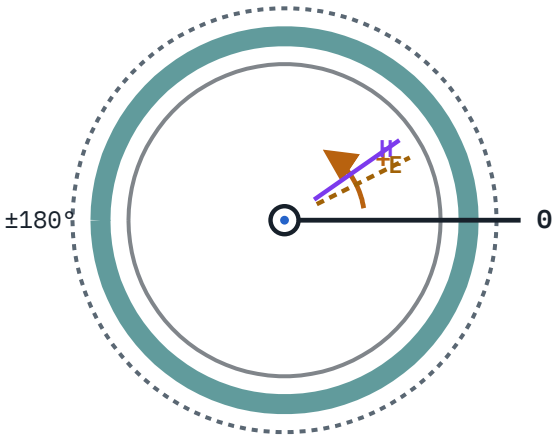
FIGURE 3 · RANGE DIALS

Limits measured on the same angle as /joint_states

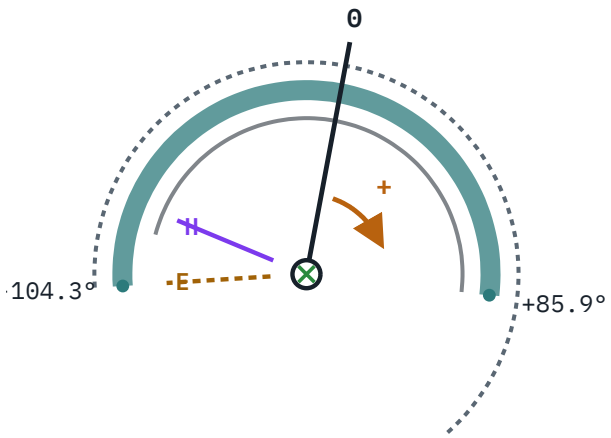
URDF limit used by MoveIt and our code — Stock ROBOTIS URDF

Servo EEPROM limit (hard stop in firmware) — Positive direction — H home — E exit / rest

joint1 · ID 11 · axis +z



joint2 · ID 12 · axis +y

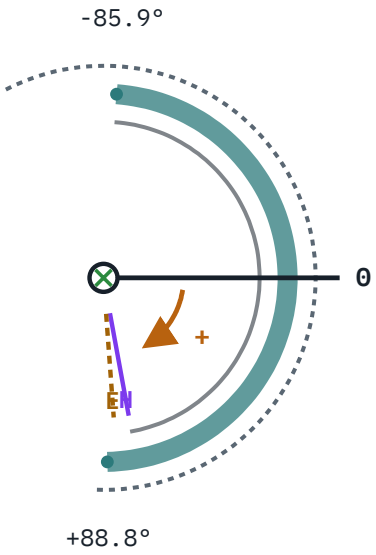


Top view: 0 = arm pointing forward (+x) — Side view: 0 = upper arm at the zero pose, which

Top view. 0 = arm pointing forward (+x).

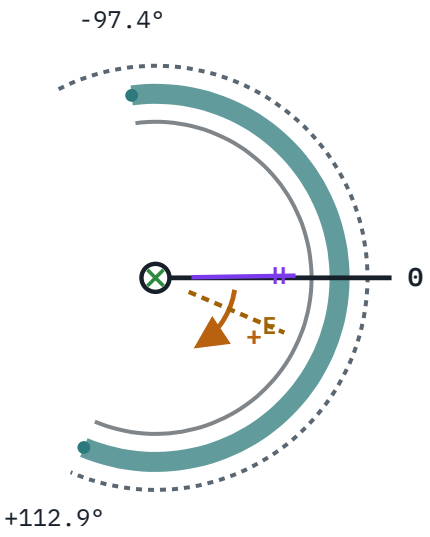
Side view. 0 = upper arm at the zero pose, which leans 10.6° forward of vertical (elbow sits 24 mm ahead). Upper arm truly vertical = −10.6°.

joint3 · ID 13 · axis +y



Side view. 0 = forearm parallel to the upper-arm frame's x (horizontal at the zero pose). Grey stub = upper arm.

joint4 · ID 14 · axis +y



Side view. 0 = gripper in line with the forearm. Grey stub = forearm.

TABLE 1

Axes, positive direction and limits per motor

JOINT	AXIS	POSITIVE ANGLE MEANS	URDF (IN USE), RAD	STOCK ROBOTIS, RAD	SE
joint1 ID 11	+z	Arm swings toward the robot's left (counter-clockwise seen from above)	-3.14 … +3.14 -180.0° … +180.0°	-3.14 … +3.14 -180.0° … +180.0°	-3 -1
joint2 ID 12	+y	Upper arm leans forward and down	-1.82 … +1.50 -104.3° … +85.9°	-1.50 … +1.50 -85.9° … +85.9°	-1 -1
joint3 ID 13	+y	Forearm swings down (elbow folds)	-1.50 … +1.55 -85.9° … +88.8°	-1.50 … +1.40 -85.9° … +80.2°	-2 -1
joint4 ID 14	+y	Gripper pitches down	-1.70 … +1.97 -97.4° … +112.9°	-1.70 … +1.97 -97.4° … +112.9°	-2 -1
gripper ID 15	+y	Prismatic, metres. gripper_left_joint moves the left finger	−0.011 … +0.020 m URDF	−0.010 … +0.019 m controller / our code	cu Op

	moves the left finger along +y; the right finger mirrors it (axis −y). Positive = open.

How the numbers fit together

- Servo to joint angle: `θ = (Present Position − 2048) × 0.001534 rad` (0.088° per tick). Present Position already includes each servo's Homing Offset (−17, 166, 38, −124 ticks for IDs 11–14), which is what puts 0 rad at the zero pose.
- The servo EEPROM limits are the firmware hard stops pinned in `open_manipulator_x_position.ros2_control.xacro`. The URDF limit is what MoveIt, `joint_commander` and `vla_bridge` respect, and must sit inside the servo range. The bridge also keeps 0.05 rad away from the URDF limit.
- We widened two URDF limits past stock ROBOTIS: joint2 lower −1.5 → −1.82 and joint3 upper 1.4 → 1.55, so the measured rest pose fits. joint2's lower URDF limit (−1.82) is 0.0007 rad past the servo stop (−1.8193); the servo stop wins.
- MoveIt reports the forearm against the shoulder bracket when θ3 is above about 1.38 rad, whatever θ2 is (known issue #12).
- The ROBOTIS e-Manual specification page gives the dimensions (reach 380 mm) and the gripper range −0.010 … 0.019 m, but no joint-angle table. The joint axes and stock limits come from the URDF ROBOTIS publishes in its `open_manipulator` repository (the initial